

NAZMUL HUDA BADHON

badhon15-4772@diu.edu.bd
[Portfolio](#) • [LinkedIn](#) • [Google Scholar](#)

+8801903906230
Dhaka, Bangladesh

SUMMARY

Final-year Computer Science undergraduate (graduating Jan 2026), specializing in Deep Learning, Medical Imaging, and Medical AI. Author of 8+ peer-reviewed publications in Q1 and Q2 journals (Elsevier, Wiley), with additional works under review. Experienced in mentoring 30+ students through research training and academic support, and motivated to advance clinically impactful AI research through interdisciplinary collaboration, with a current focus on securing a Research Assistantship position.

EDUCATION

BSc	Daffodil International University, Computer Science and Engineering CGPA: 3.88/4.00	Spring 2022–Fall 2025 Dhaka, Bangladesh
	Exchange Semester: Dongseo University, Computer Engineering GPA: 3.94/4.50, Obtained Marks: 93.25% in total	Fall 2023 Busan, South Korea

RESEARCH INTERESTS

Deep Learning, Machine Learning, Medical Imaging, Medical AI.

RESEARCH PUBLICATIONS

Journal Publication

- Badhon NH**, Salehin I, Sajib MTA, Rifat MHS, Noman SM, Moon NN. (2025). “TLCNN: Tabular data-based Lightweight Convolutional Neural Network for Electricity Energy Demand Prediction.” *Global Energy Interconnection* (Elsevier). [Accepted \[Q1\]](#) (in press).
- Salehin I, **Badhon NH**, Sajib MTA, Moon NN. (2025). “Secure Medical AI: Federated Learning and Large Language Models for Medical Intelligence.” *Engineering Reports* (Wiley). [Submitted \[Q2\]](#)
- Sajib MTA, **Badhon NH**, Salehin I, Rifat MSR, Ahmmed F, Moon NN. (2025). “A comparative deep learning methodology for plant insect image classification: assessment of CNN architectures and augmentation techniques.” *MethodsX* (Elsevier). [Under Review \[Q2\]](#)
- Salehin I, Sajib MTA, **Badhon NH**, Rifat MHS, Amin N, Moon NN. (2025). “Systematic Literature Review of LLM in Medical: Digital Health, Technology and Applications.” *Engineering Reports* (Wiley). [\[Q2\]](#). [DOI Link](#).
- Salehin I, **Badhon NH**, Sajib MTA, Moon NN. (2025). “DeepUS-ReconSeg: A Multi-Angle Paired B-Mode Ultrasound Dataset for Medical Imaging Reconstruction and Segmentation.” *Data in Brief* (Elsevier). [DOI Link](#).
- Salehin I, **Badhon NH**, Sajib MTA, Saifuzzaman M, Moon NN. (2025). “IQ-UltraRecon: Demodulated IQ Ultrasound Dataset of Human Hand and Arm Tissue for Deep Learning-Based Reconstruction.” *Data in Brief* (Elsevier). [DOI Link](#).
- Salehin I, Khan MR, Habiba U, **Badhon NH**, Moon NN. (2024). “BAU-Insectv2: An Agricultural Plant Insect Dataset for Deep Learning and Biomedical Image Analysis.” *Data in Brief* (Elsevier). [DOI Link](#).

Book Chapter

- Badhon NH**, Salehin I, Sajib MTA, Rifat MHS, Moon NN. (2025). “Predicting Peak Energy Demand Using Machine Learning Techniques for Efficient Electricity Supply Management.” In *Engineering Applications of AI for Demand Forecasting* (Taylor & Francis). [Accepted](#) (in press).

PROJECTS

- Breast Cancer: Segmentation-Guided Hybrid Classification**, [Link](#).
Built a two-stage U-Net and CNN-based hybrid model on BUSI datasets for lesion segmentation and classification, achieving strong performance on accuracy and ROC metrics.
- Lung–Colon Histopathology: Image Enhancement & Transfer Learning**, [Link](#).
Applied preprocessing techniques on histopathology images and evaluated multiple transfer learning models, identifying the best-performing model by accuracy and F1-score.
- Medical Image Preprocessing & Quality Analysis with Vision Transformer**, [Link](#).
Designed preprocessing workflows for medical images and classified with ViT, achieving **87% testing accuracy** and improved transformer performance.

TECHNICAL SKILLS

Programming: Python, C, Java

Deep Learning Algorithm: Intermediate Level

Machine Learning Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV

Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn

Tools: GitHub, Google Colab, LaTeX

TEACHING & MENTORSHIP EXPERIENCE

- Conducted 10+ research training sessions for 30+ student participants at the Multidisciplinary Action Lab, Daffodil International University.
- Assisted in delivering undergraduate academic content in Data Science, Database, and Big Data under faculty supervision.

AWARDS & LEADERSHIP

Academic

- Awarded a fully funded [Global Korea Scholarship \(GKS\)](#) for Exchange Study at Dongseo University by the Government of the Republic of Korea. *Fall 2023*
- Received a [Special Scholarship](#) for academic results from Daffodil International University. *Spring 2022–Fall 2025*

Research

- Received a [Research Award](#) from the Division of Research, Daffodil International University. *2024*
- Served as a [peer reviewer](#) for *Digital Health* (SAGE Publishing) and received an official reviewer certificate. *2025*

Leadership

- Led a team of [20+ students](#) in International Camping at Dongseo University, South Korea. *2023*
- Completed the [Volunteer Internship Program](#) under the International Affairs, Daffodil International University. *2024*

REFERENCES

1. Nazmun Nessa Moon

Associate Professor,
Department of Computer Science and Engineering,
Daffodil International University, Bangladesh
Email: moon@daffodilvarsity.edu.bd

2. Imrus Salehin

Research Assistant, AI-based Medical Imaging Lab,
Department of Information Convergence Engineering,
Pusan National University, South Korea.
Email: salehin@pusan.ac.kr